Xi'an Jiaotong-Liverpool University

西交利物浦大學

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT202			

2nd SEMESTER 2023/24 FINAL EXAMINATION

UNDERGRADUATE – YEAR 3

COMPLEXITY OF ALGORITHMS

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. **This is a closed-book examination.**
2. **The examination will be conducted in all other respects in accordance with the Regulations for the Conduct of Examinations.**
3. **Total marks available are 100. This exam will count for 80% of the final mark.**
4. **Answer all questions. There is NO penalty for providing a wrong answer.**
5. **Only English solutions are accepted.**

Question 1. [10+10=20 MARKS]

A) Consider the following code fragment.

```
function mystery(n)
    r=0
    for i=1 to n-1 do
        for j=i+1 to n do
            for k=1 to j do
                output foobar
    return(r)
```

Let T(n) denote the number of times `foobar' is printed as a function of n.

1) Express T(n) as a summation (actually two nested summations). **[5 marks]**
2) Simplify the summation, and give the worst-case running time using Big-O notation. **[5 marks]**

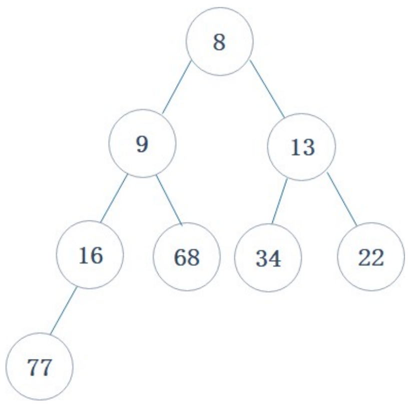
B) We have the following piece of code, representing a recursive function.

```
function NoIdea (n)
   if n > 1:
      print `A'
       NoIdea (n/3)
           for i = 1 . . . n
               print `B'
           end for
        NoIdea (n/3)
```

1) What is the runtime of the above function? Express your answer using the Big-O notation. **[5 marks]**
2) Express the number of times that this algorithm prints `A' in terms of n using the Big-O notation. **[5 marks]**

Question 2. [10 MARKS]

Delete two minimum numbers on the following min-heap. You do not need to show the array representation of the heap. You are only required to draw intermediate heaps and circle the final step. **[10 marks]**



Question 3. [8 MARKS]

Let set S = {a, b, c, d, e} denote a set of objects with weights and benefits as given in the table below.

Item	a	b	c	d	e
Benefit	13	9	8	15	6
Weight	5	5	5	3	1

What is an optimal solution to the fractional Knapsack problem for S assuming that we have a sack that can hold objects with total weight 18? **[8 marks]**

Question 4. [6+6=12 MARKS]

A) Prove that

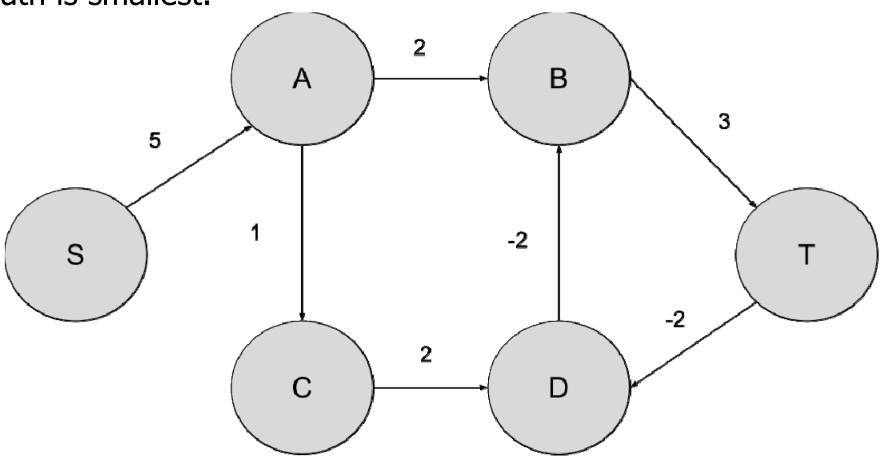
$$logn + loglog\, n \text{ is } \Theta(logn).$$ **[6 marks]**

B) Give a tight upper bound (Big-O notation) on the solution to the following function. You need to justify your answers. **[6 marks]**

$$T(n) = \begin{cases} 1, & if\ n = 1 \\ T(n-1) + n(n-2), & if\ n \geq 2 \end{cases}$$

Question 5. [10+5=15 MARKS]

For a given weighted directed graph and two nodes s and t, the shortest path problem is to find a simple path (without repeating nodes) from S to T such that the sum of weighted edges in the path is smallest.



(1) Give a step by step process of using Dijkstra's algorithm for finding a shortest simple path from S to T in the given graph. **[10 marks]**

(2) What is the shortest simple path and its' value? **[5 marks]**

Question 6. [5+5+5=15 MARKS]

(1) Show $24^{2401} \ mod \ 2501 = 24$ using Euler's theorem. **[5 marks]**

Alice and Bob wish to use the RSA cryptosystem for communication. Alice announced the public modulus and encryption key pair to be (2501, 343).

(2) Derive the private key. **[5 marks]**

(3) Assume Bob sent an encrypted message 4, what's the original unencrypted message? **[5 marks]**

Question 7. [5+10+5=20 MARKS]

A *Clique* C of an undirected graph G = (V, E) is a subgraph of G such that all pairs of two vertices in C are adjacent to each other. In other words, it is a complete subgraph.
The Clique decision problem asks "Given an undirected graph G = (V,E) and a positive integer k, and the problem is to check whether a clique of size k exists in G."

(1) Is the Clique decision problem in NP? Prove or disprove it. **[5 marks]**

(2) Provide a reduction from the 3SAT problem to the Clique decision problem, and prove its' correctness. **[10 marks]**

(3) Prove or disprove that Clique decision problem is NP-complete. **[5 marks]**

END OF EXAM PAPER